

AIX

Linear Regression / Logistic Regression

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Mar, 5th, 2026

Linear and Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Goal framing:
 - Architecture idea
 - Data & training paradigm → expand to domain

**Least Squares
(1805)**

measurement error

Linear Regression

$$\text{score} = \mathbf{w}^T \mathbf{x} + b$$

Logistic Regression

$$\text{prob} = \sigma(\text{score})$$

Perceptron → MLP

non-linearity
+ stacking

Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Data = Points in Space
- Dataset

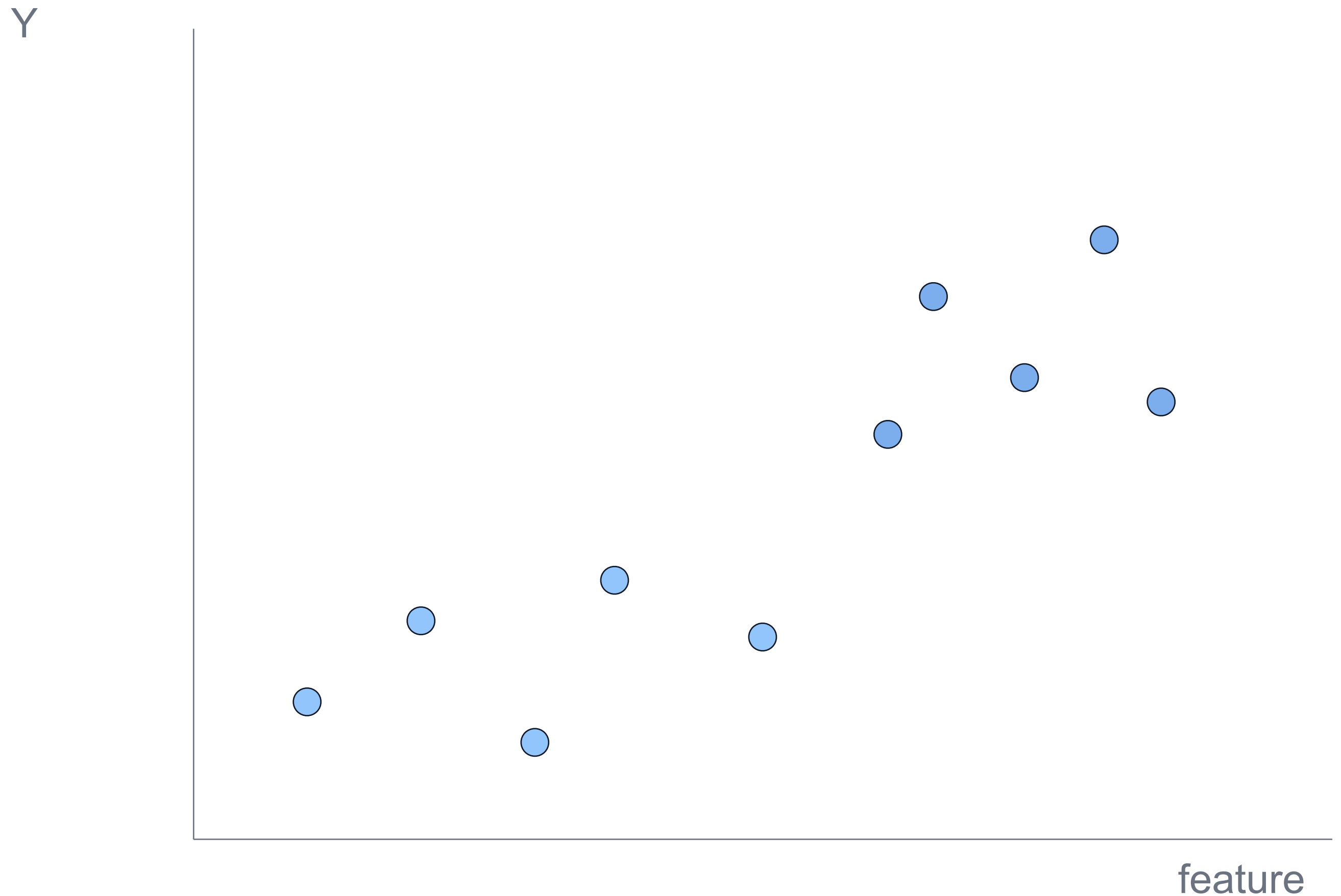
x: features

y: target

Supervised learning:

learn

$$f(x) \approx y$$



Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

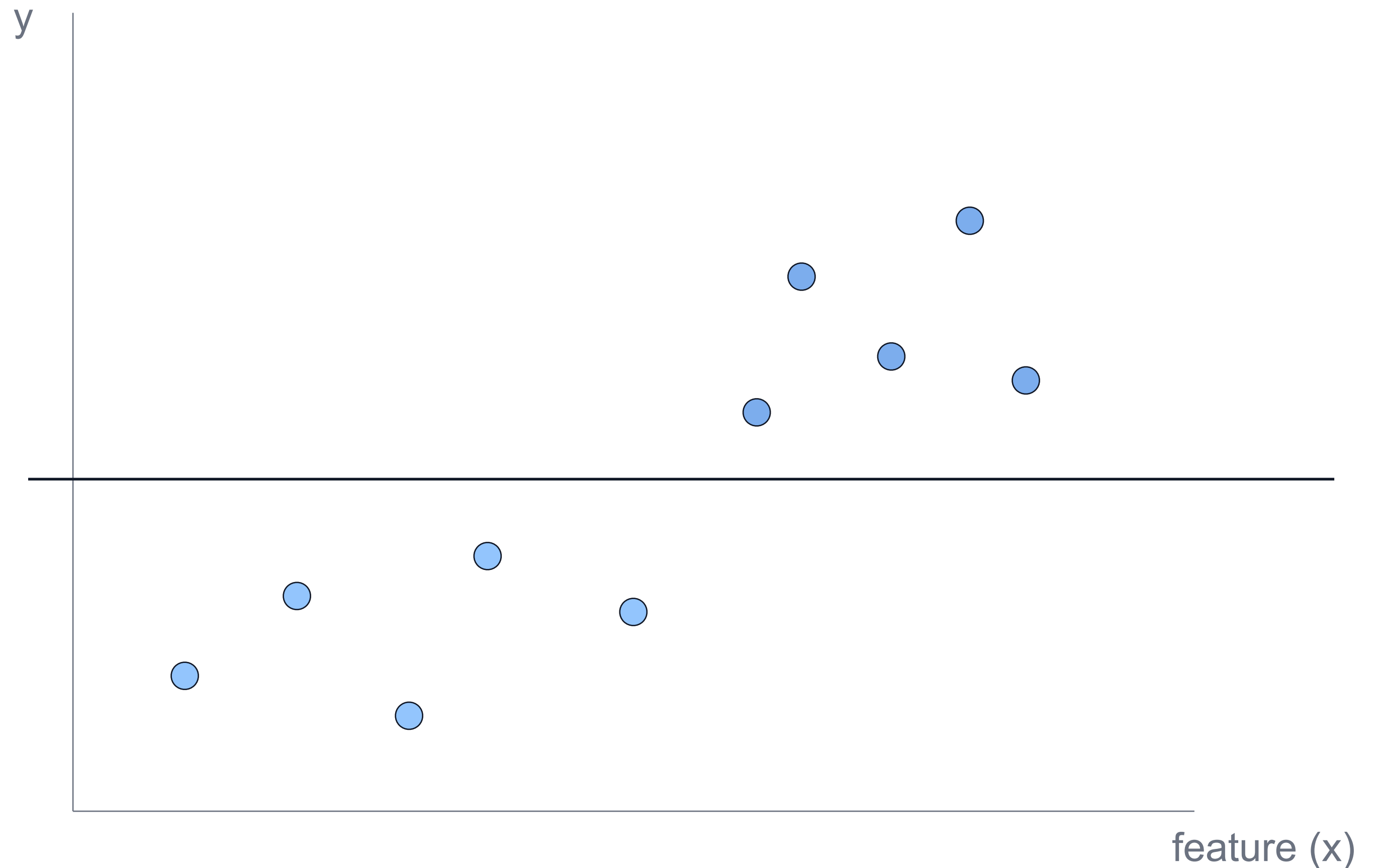
- Linear Models
 - A line / A plane

$$\text{score} = w^T x + b := \hat{y}$$

Decision boundary:

$$\text{score} = 0$$

What if $w = 0$?



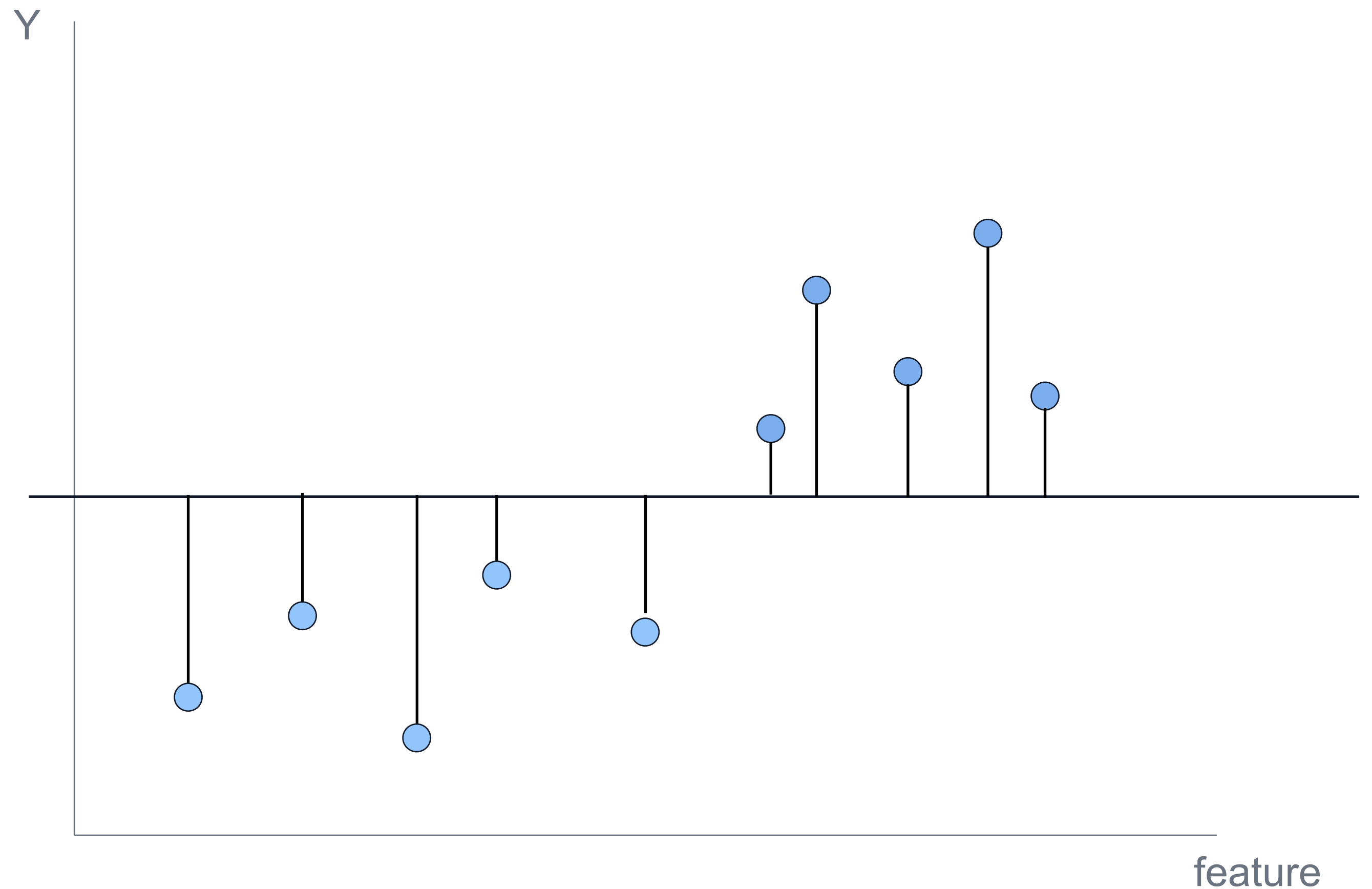
Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Residual : Orthogonal Distance

- Residual : $(y - \hat{y})$
- Squared Residual
(L2-distance) : $(y - \hat{y})^2$

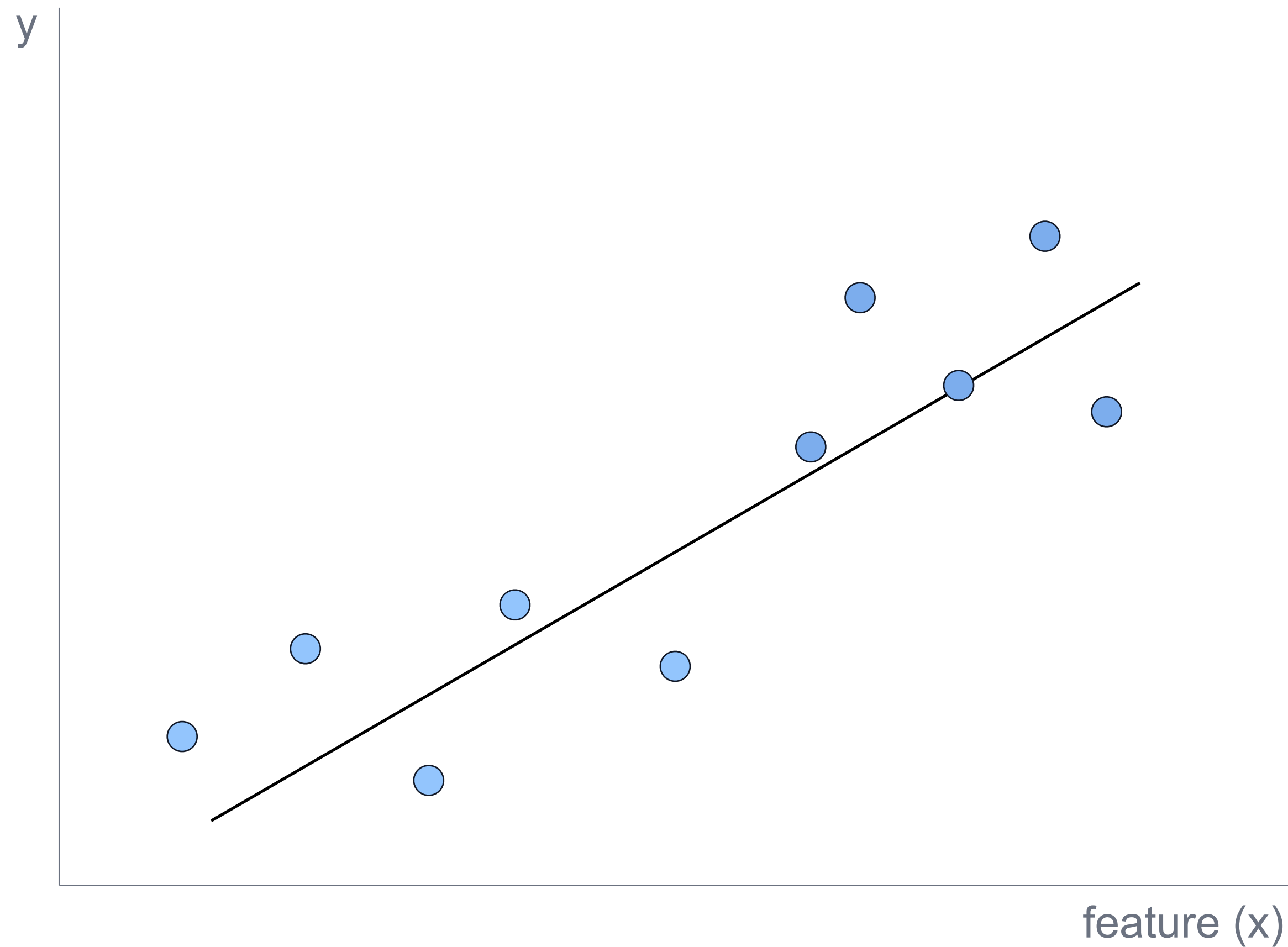
$$\text{Residual} = (y - \hat{y})$$



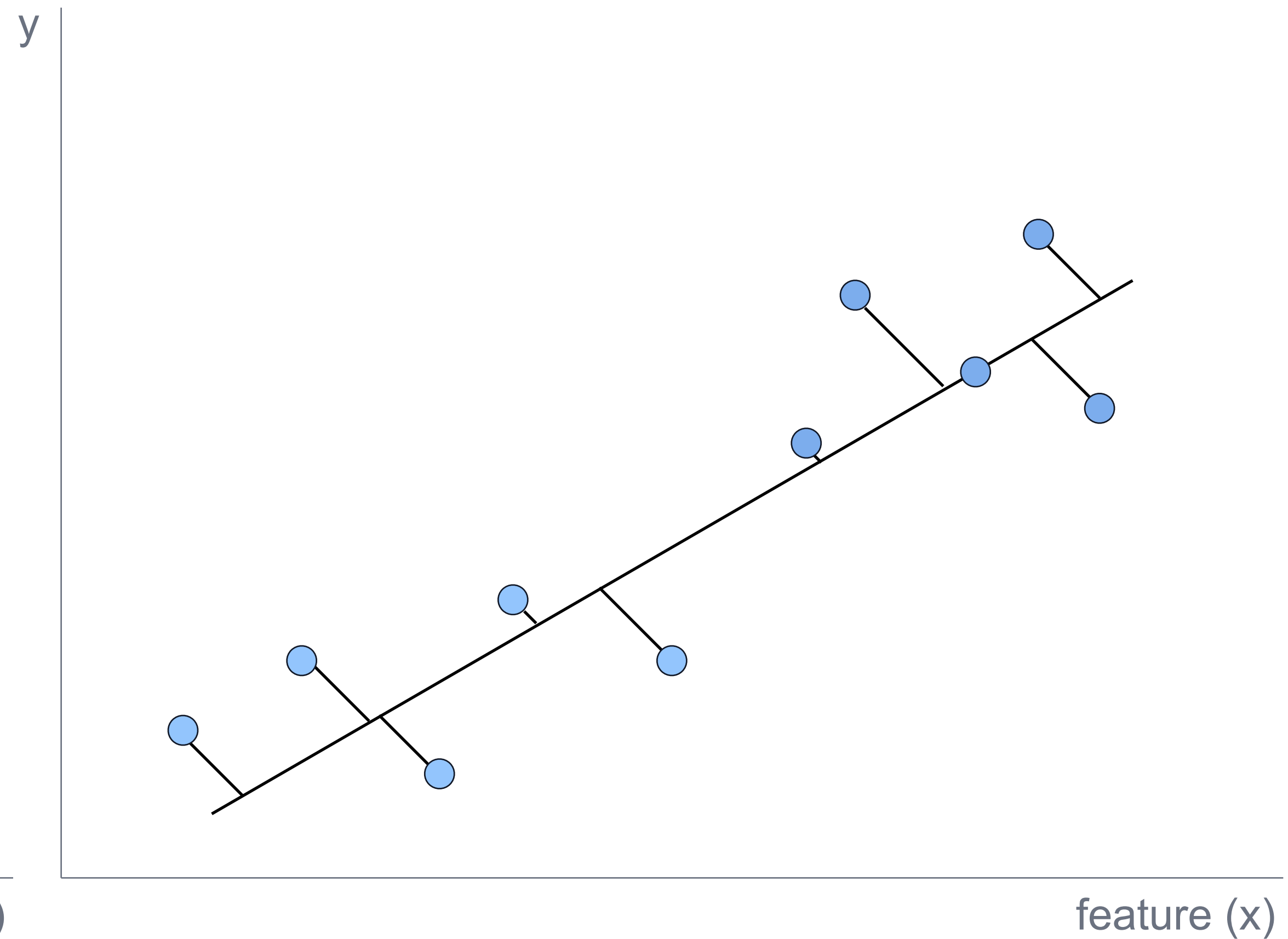
Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- What if $w = 1$?



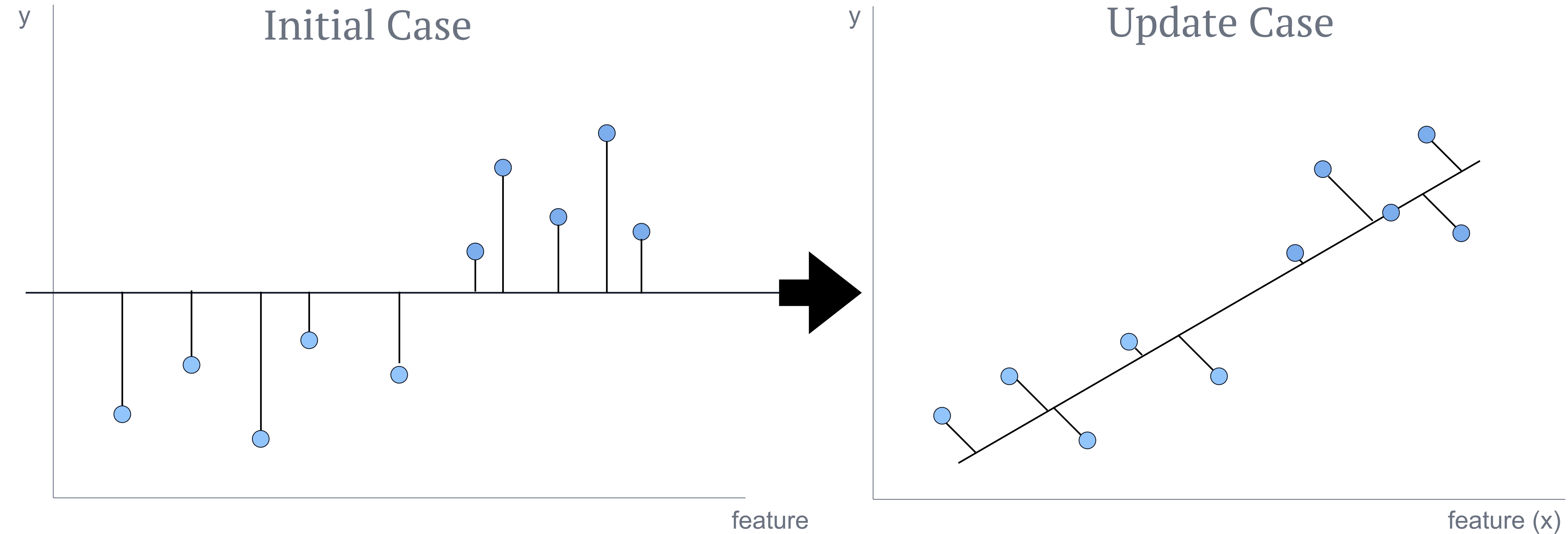
$$\text{Residual} = (y - \hat{y})$$



Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- What we want?



Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Objective Function $\approx$ Loss function
 - What difference?
 1. Residual $\sum (y - \hat{y})$
 2. Squared Sum of Residual $\sum (y - \hat{y})^2$
- Which one is suitable?

Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- How update?
 - Objective functions
 - Minimized Squared Error

$$\text{Loss} = L(w) = \sum (y_i - \hat{y}_i)^2$$

Training = choose w to minimize L



Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- How update?

Closed-form

$$w^* = (X^T X)^{-1} X^T y$$

Pseudo Inverse :

Great for small problems
(but matrix inversion can be heavy)

Iterative

$$w \leftarrow w - \alpha \nabla L(w)$$

Scales to deep learning
(we'll use this all semester)

Linear Regression

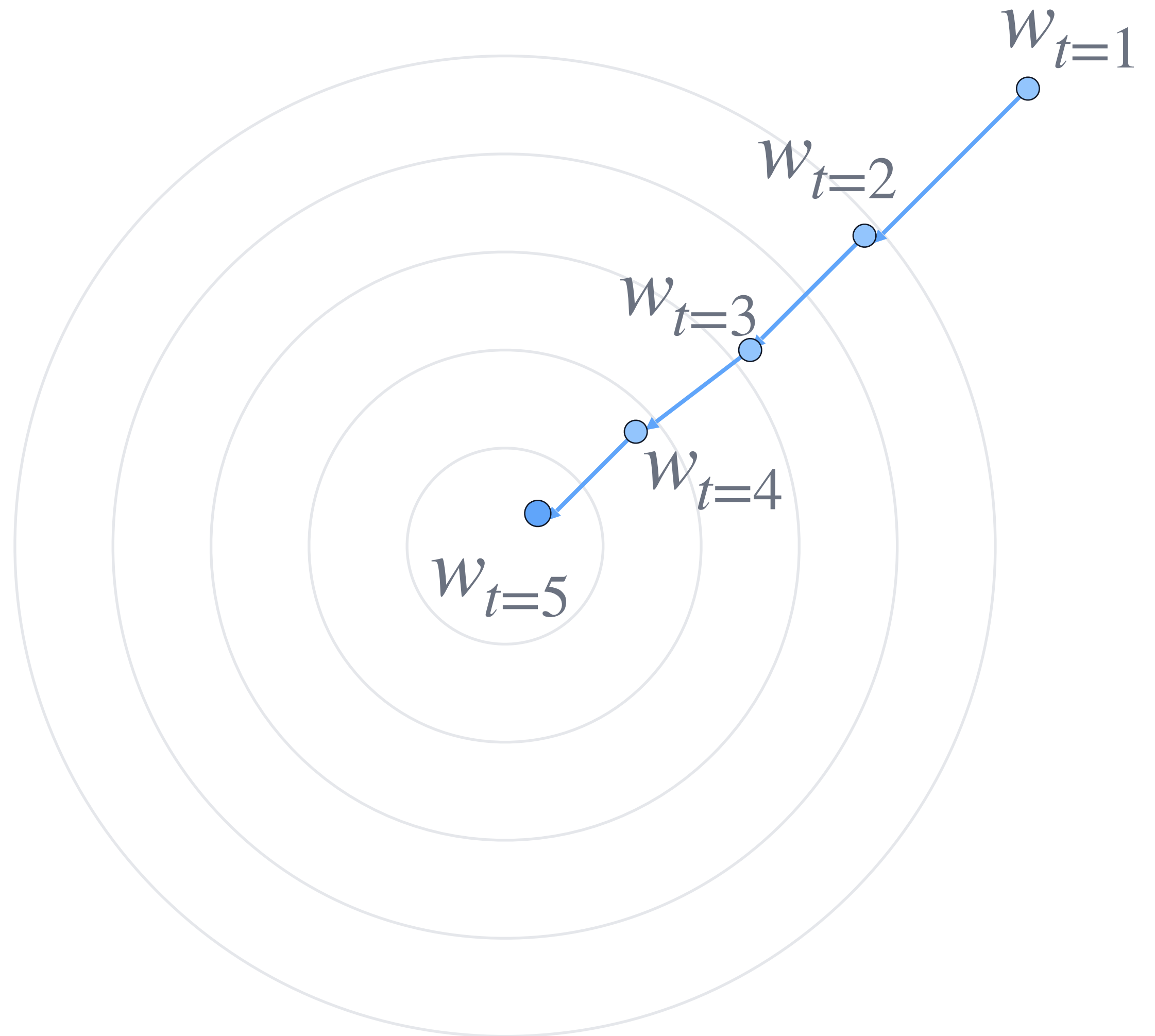
Goal: make Week 1 feel obvious before we touch MLP.

- Gradient Descent (Intuition)

Update rule:

$$W \leftarrow W - \alpha \nabla L(W)$$

In deep learning, this is “training”.



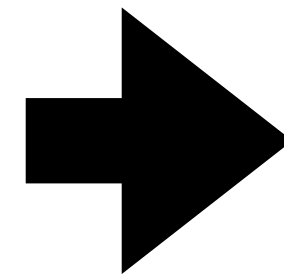
Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Before Deep Learning : Feature Engineering



Linear fit
(Under-fit)



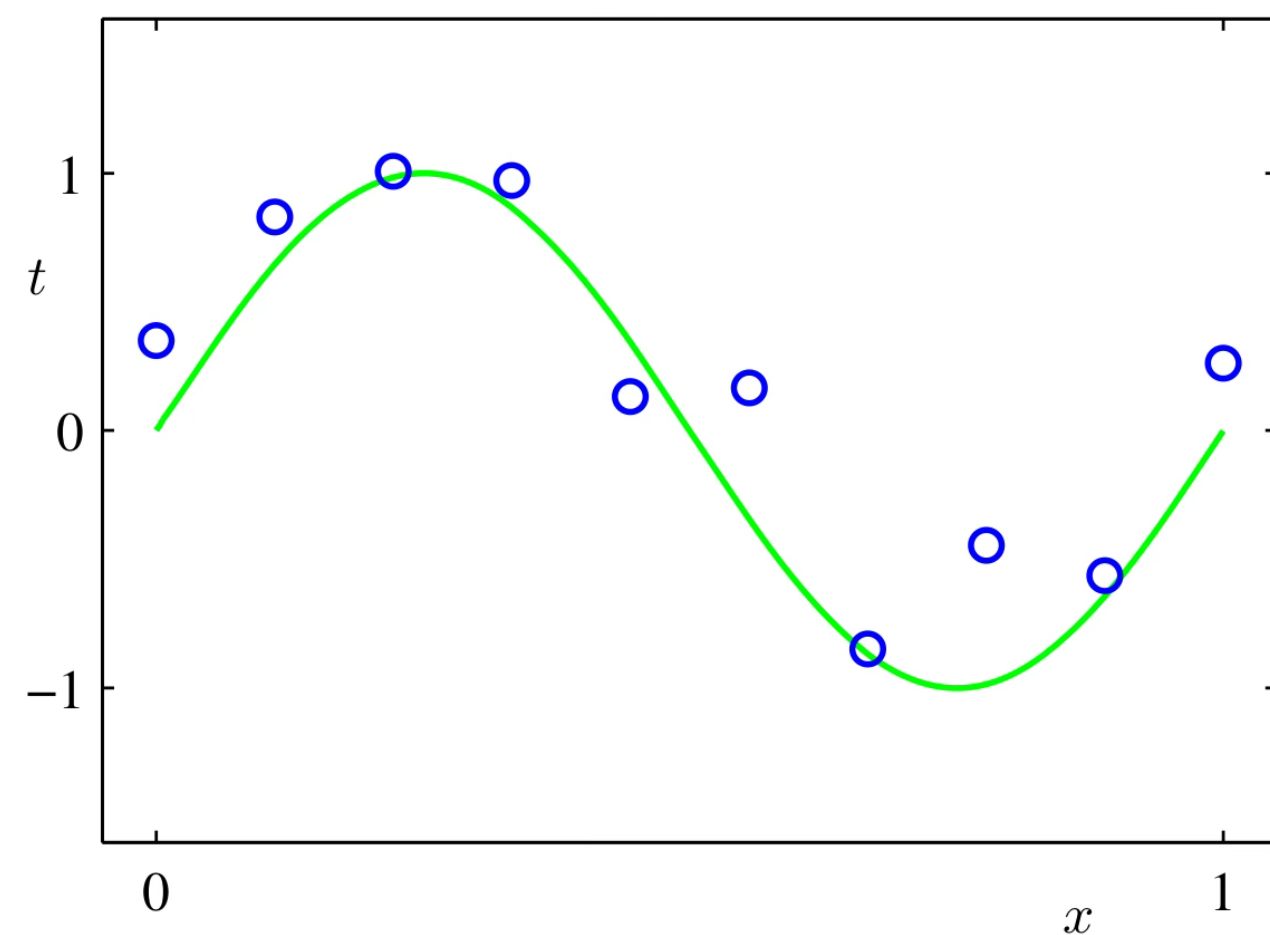
Linear + engineered features
(Improve expressiveness)

Linear Regression

Goal: make Week 1 feel obvious before we touch MLP.

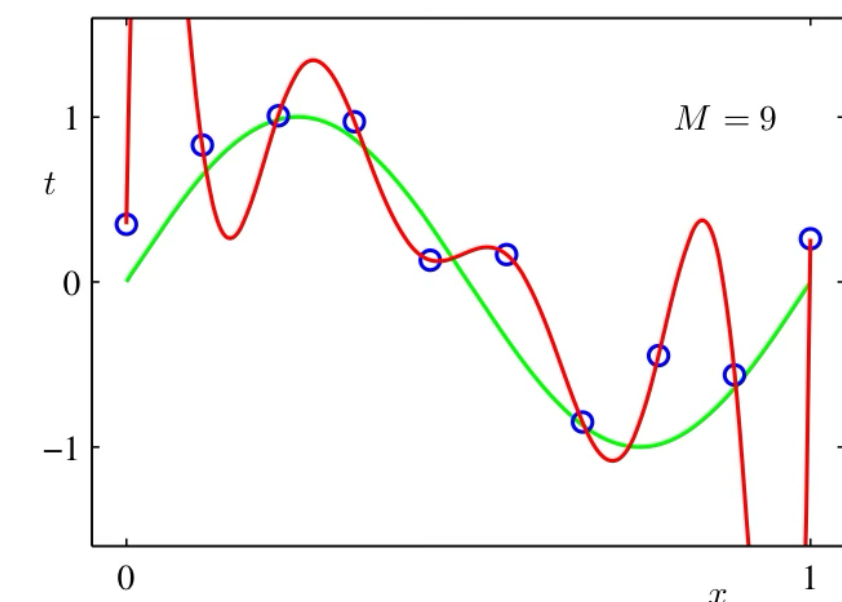
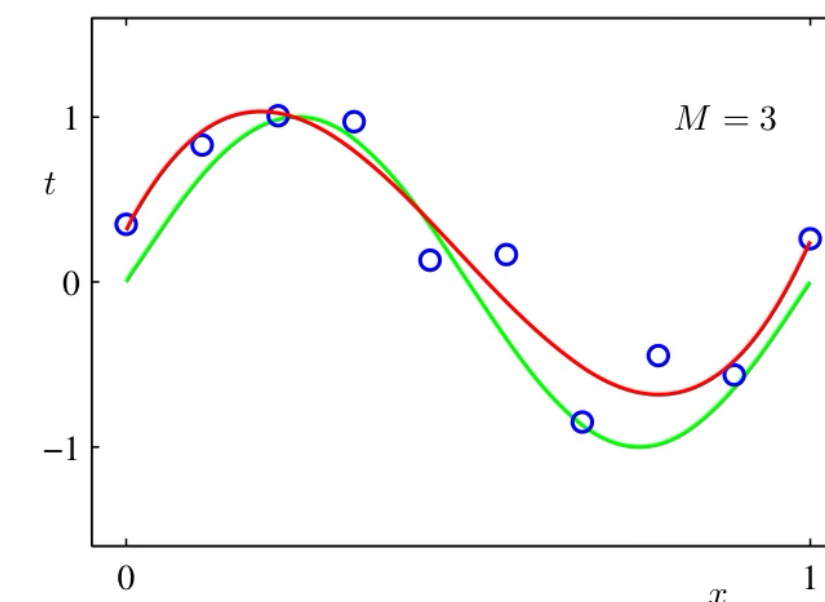
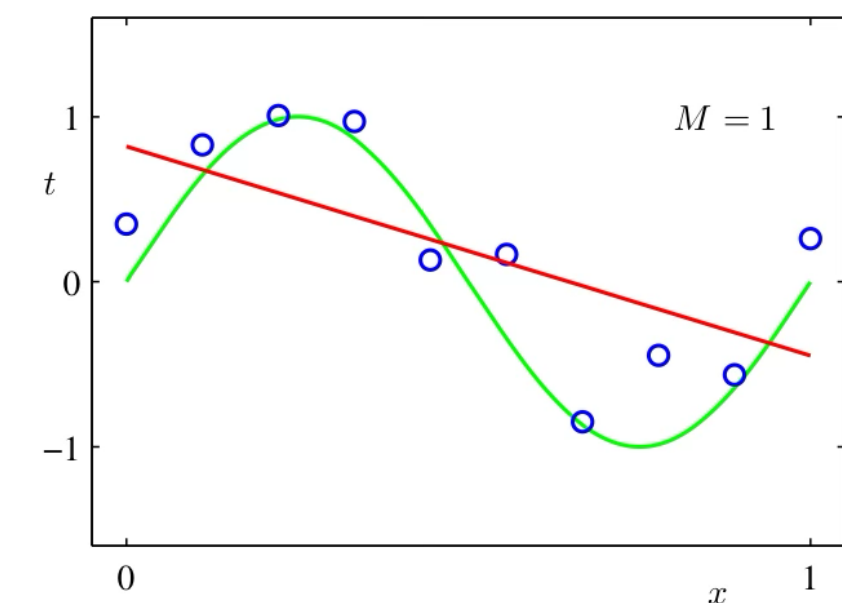
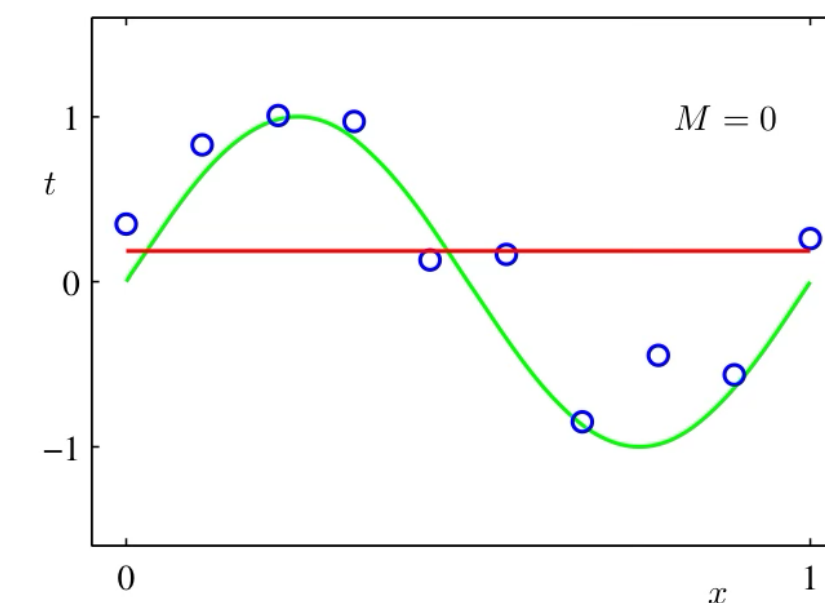
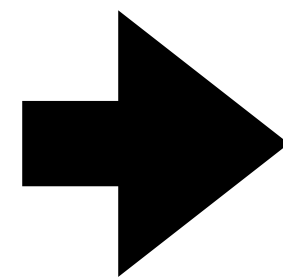
- Before Deep Learning : Feature Engineering

- Models : $y = wx + w_1x + w_2x^2 + \dots w_mx^m = \sum_{j=0}^M w_jx^j$



Ground-Truth : $\sin(2\pi x)$

Data Generation : $\sin(2\pi x) + \epsilon$

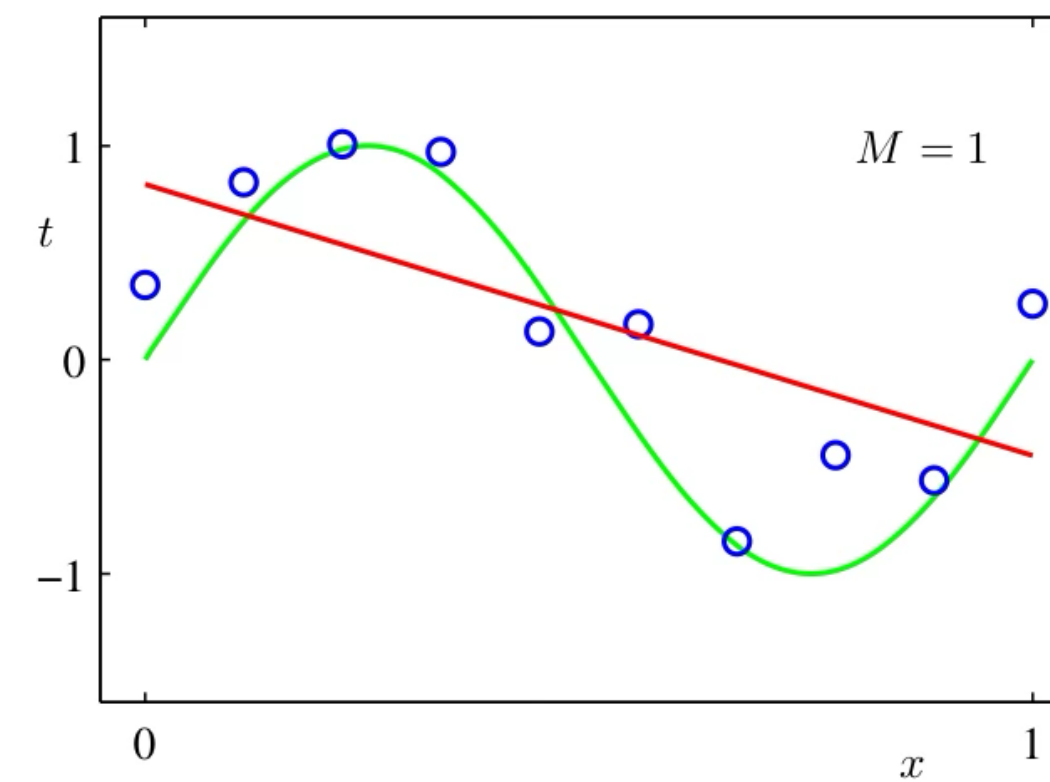
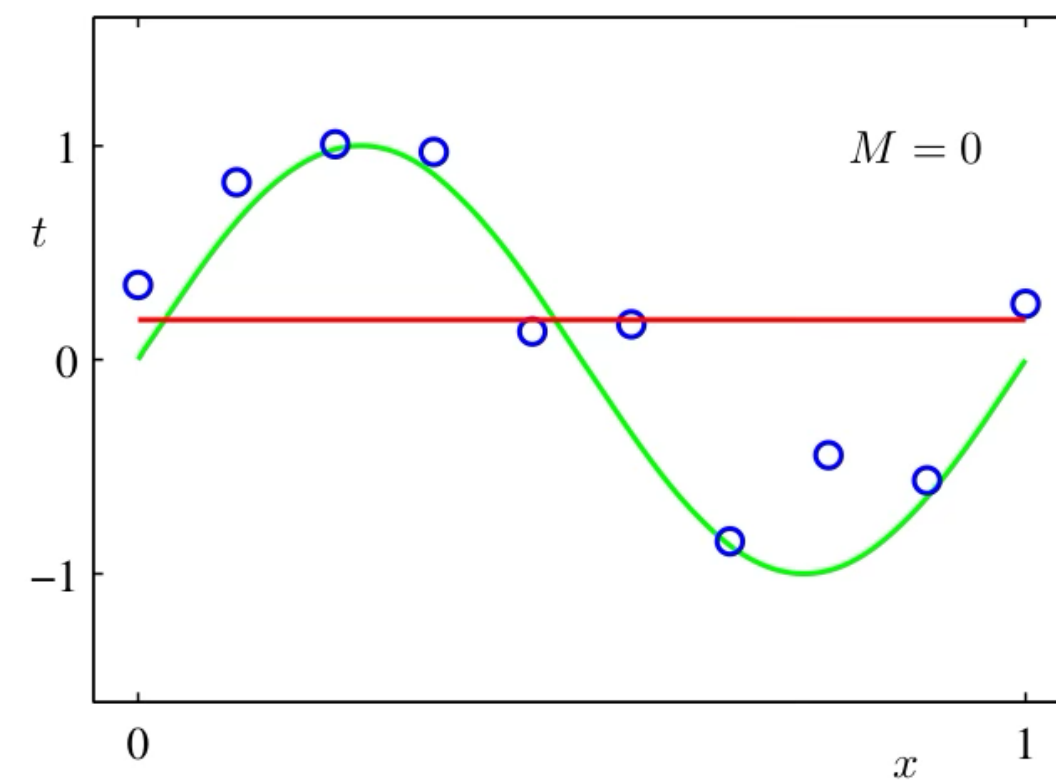


Linear Regression

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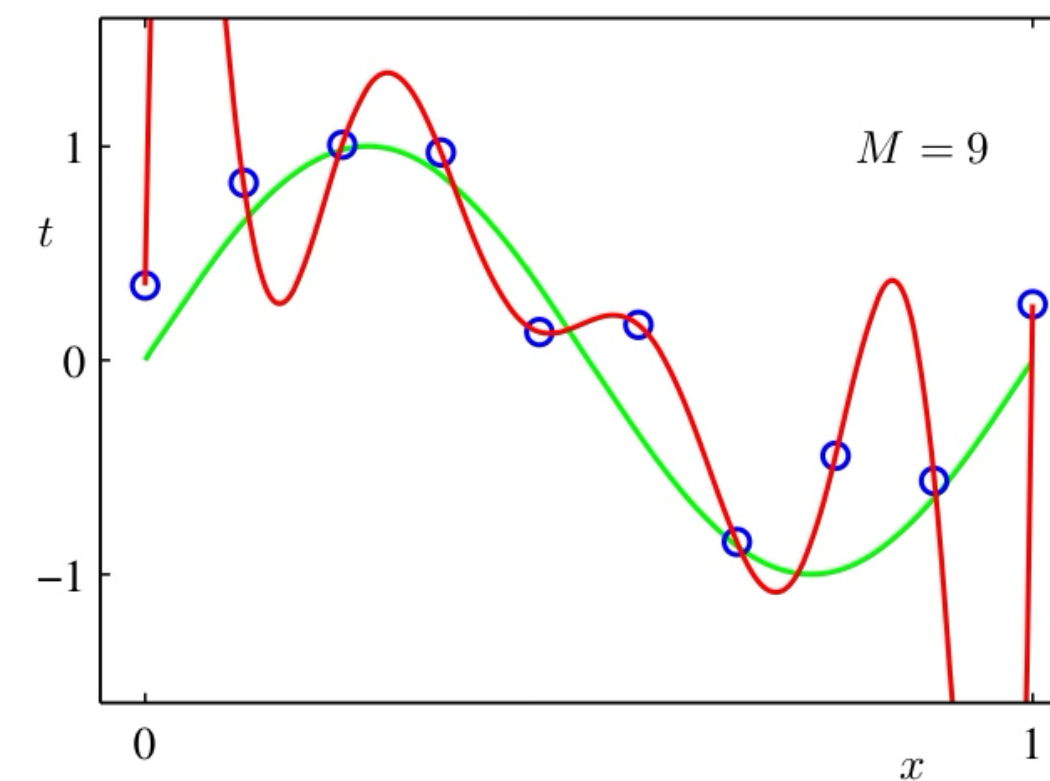
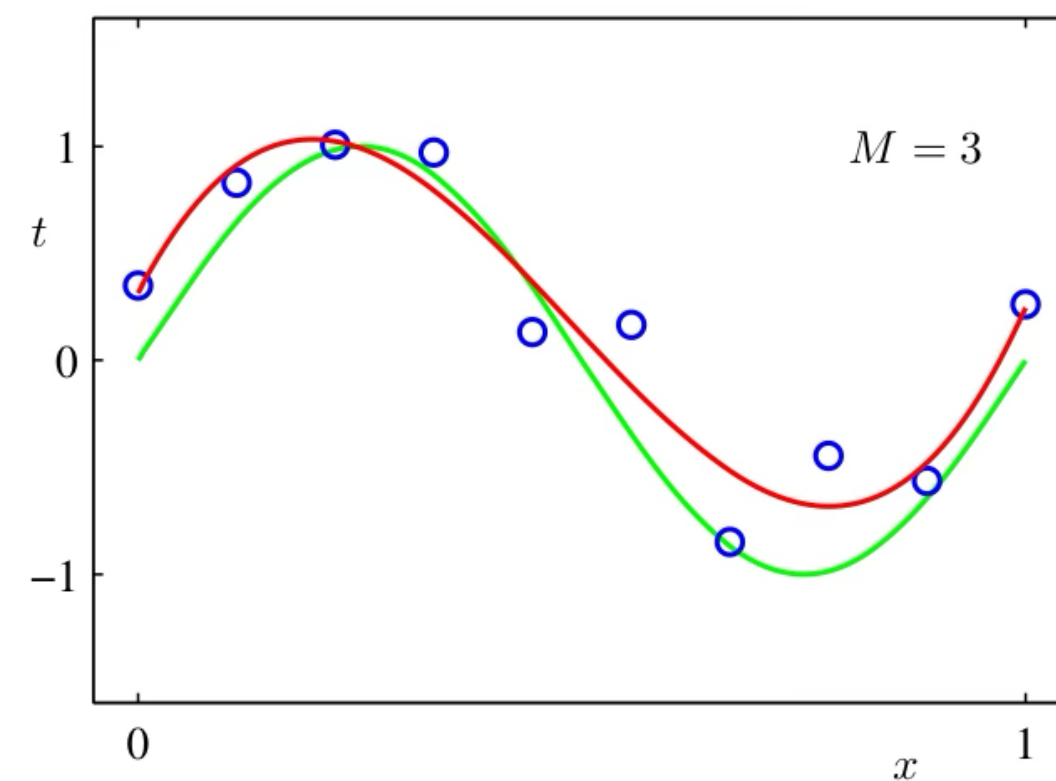
- Before Deep Learning : Feature Engineering

Under-fit



Under-fit

Good !



Over-fit

How we approach
“Classification” ?

Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- We want
 - Class probability

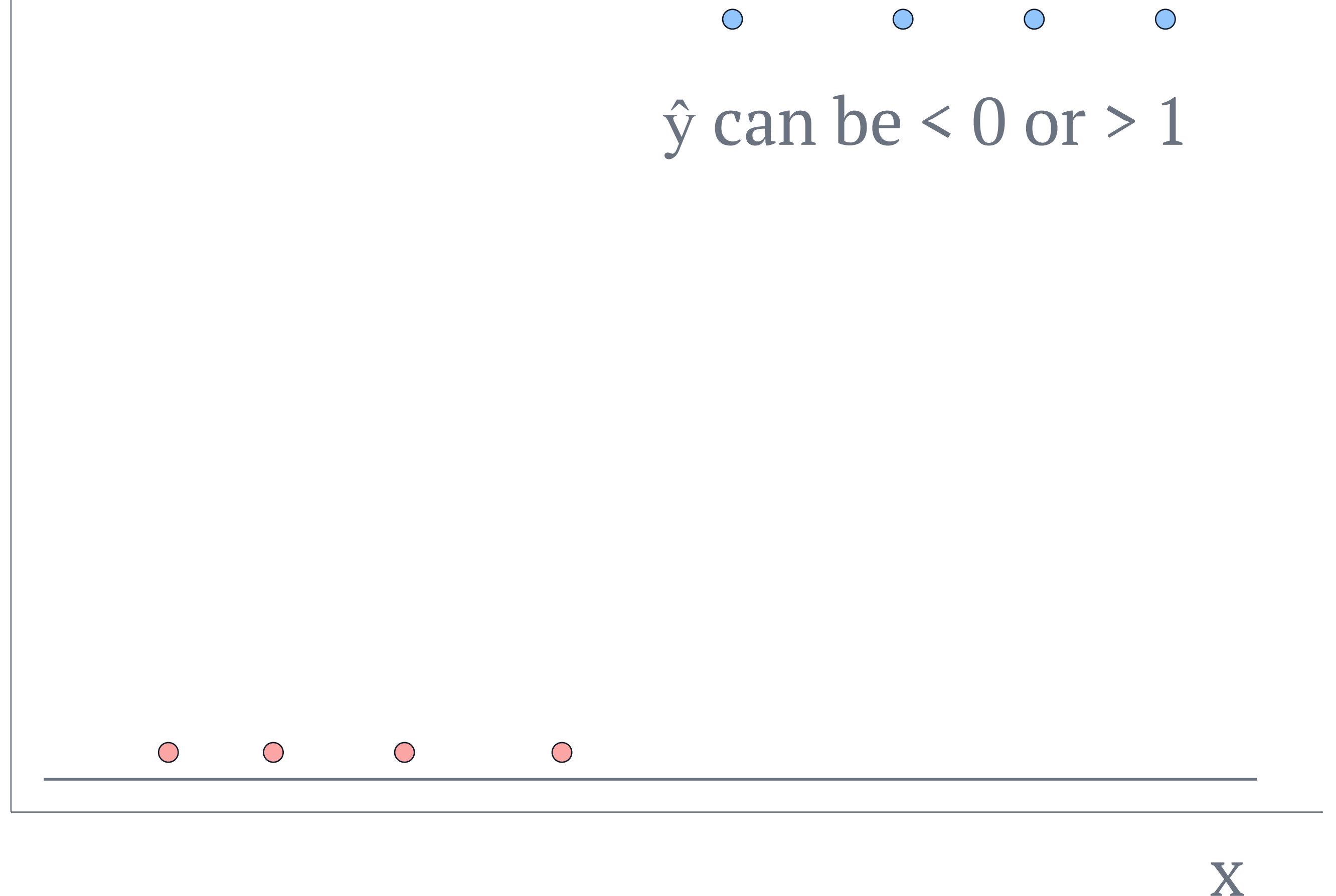
probability :

$$P(y=1|x) \in [0,1]$$

→ enter logistic regression

$$y \in \{0,1\}$$

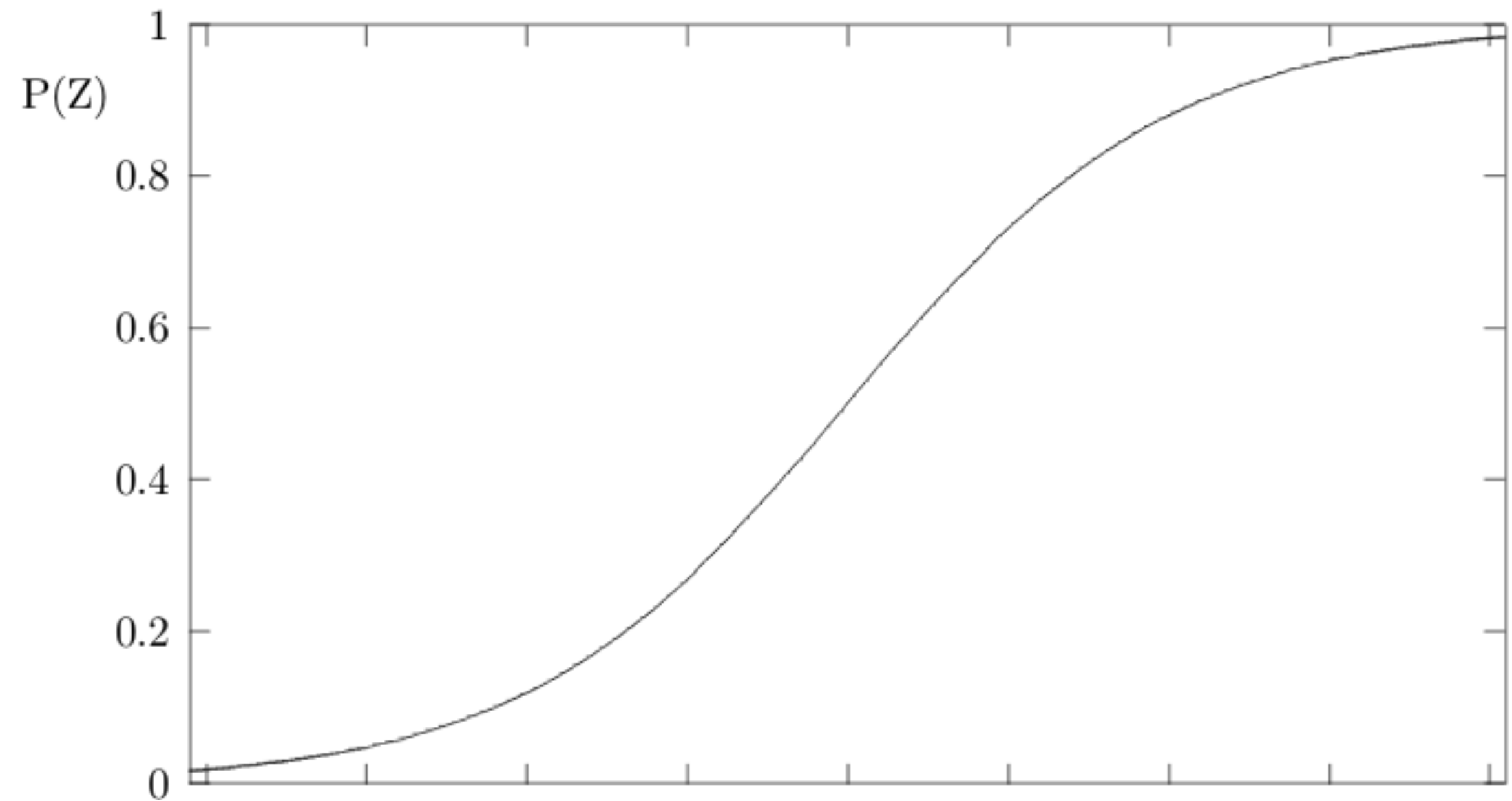
$\hat{y}$ can be < 0 or > 1



Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Logistic Curve
 - Probability from a score
 - History Cue
 - Growth Curve
 - Bioassay (생물학 실험)
 - dose $\rightarrow P(\text{response})$



Link Function : $Z \rightarrow P(\text{response})$

$\rightarrow$ logistic regression

[Sources]

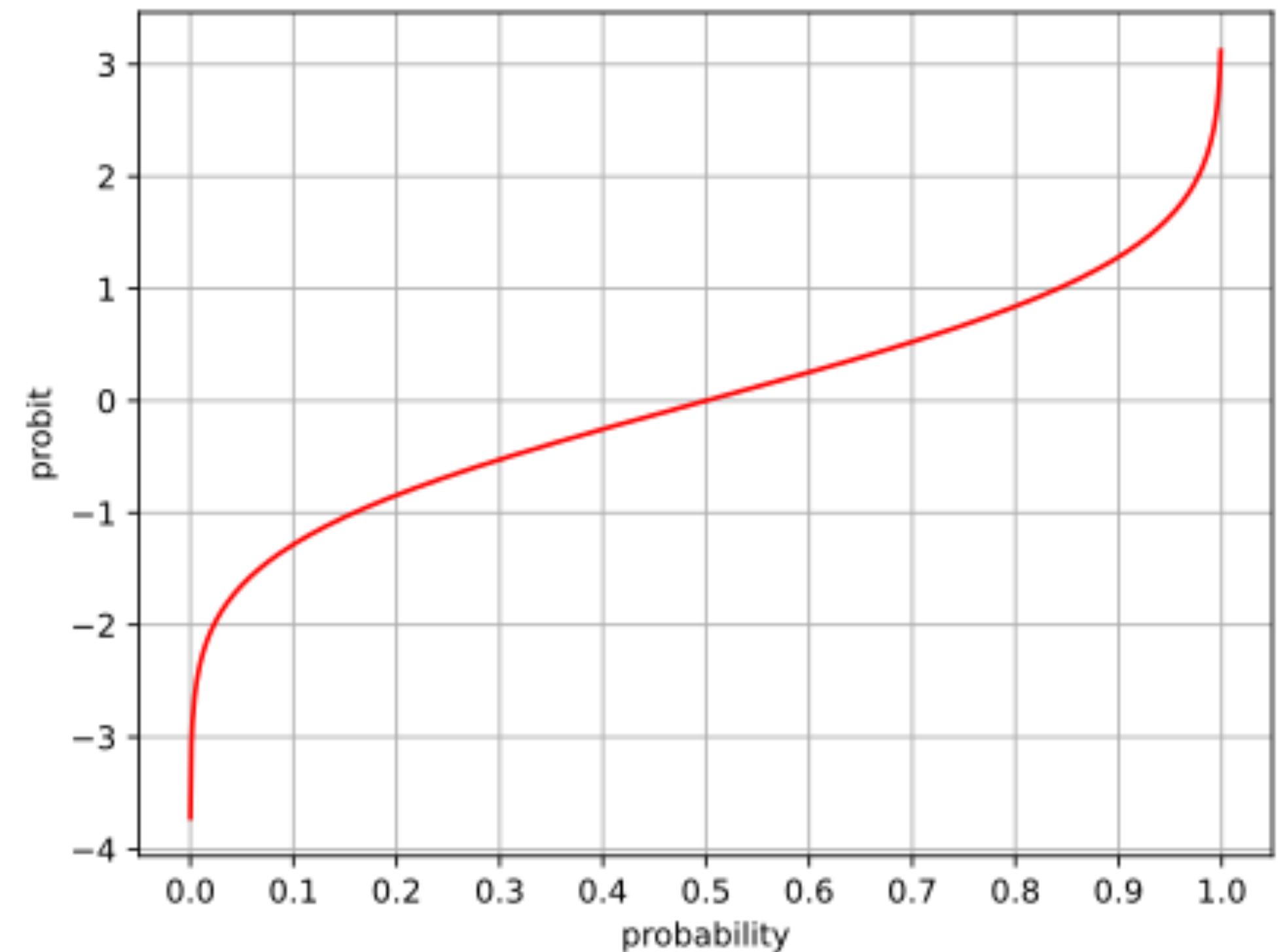
- <https://www.jhanley.biostat.mcgill.ca/anniversaries/ByTopic/OriginsLogisticRegression.pdf> (Cramer, 2002; Figure 1 logistic curve visual excerpt used).

- <https://www.tandfonline.com/doi/abs/10.1080/01621459.1944.10500699> (Berkson, 1944; example of early bioassay adoption, bibliographic reference).

Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

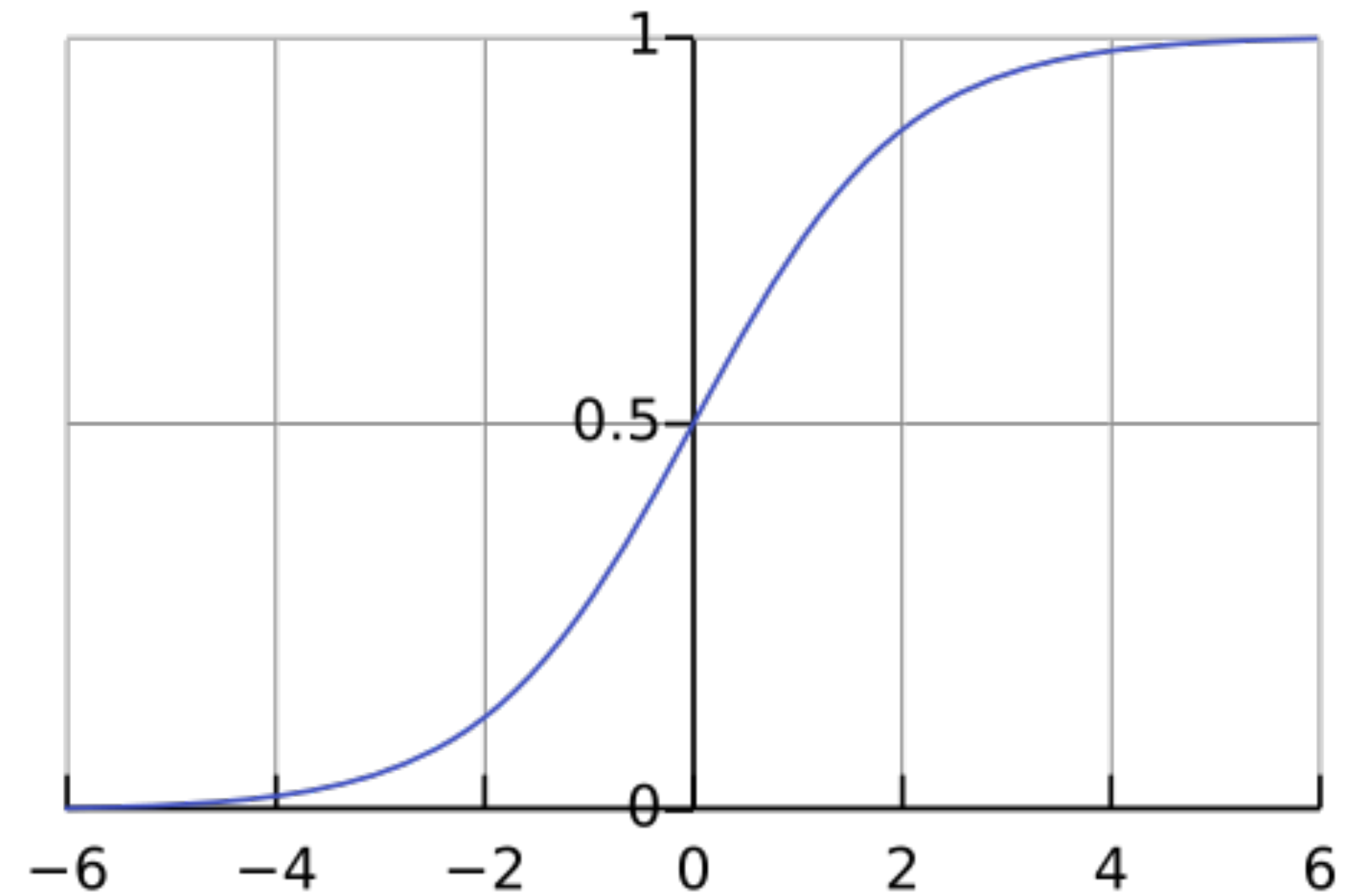
- Mathematical Expression 1
 - Growth Curve → Profit Function
 - $\Phi(Z) = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$
 - CDF (Cumulative Distribution) of Standard Normal Distribution



Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

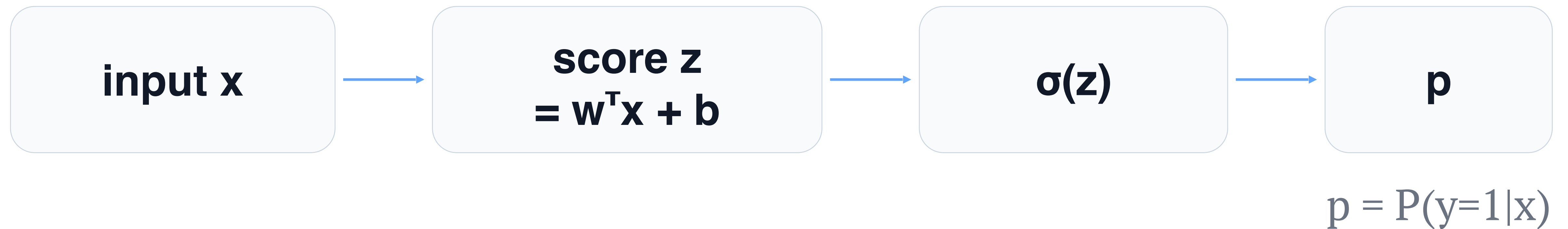
- Mathematical Expression 2
 - Sigmoid Function (S-Curve)
 - $P(Z) = \frac{1}{1 + e^{-Z}}$
 - More simple,
More easy to calculate gradient w.r.t Z
- TMI) Here, Z equals to Score from Linear Models (discuss later)



Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Architecture
 - Same “Linear Score” as linear regression
 - However, we use a probability link function



Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Training Logistic Regression
 - Loss function : Log-Likelihood

Intuition



Choose parameters so that
observed labels are “most likely”.

maximize
“ $\log p(\text{data} \mid w)$ ”

Binary cross-entropy

$$L = - [y \log p + (1 - y) \log(1 - p)]$$

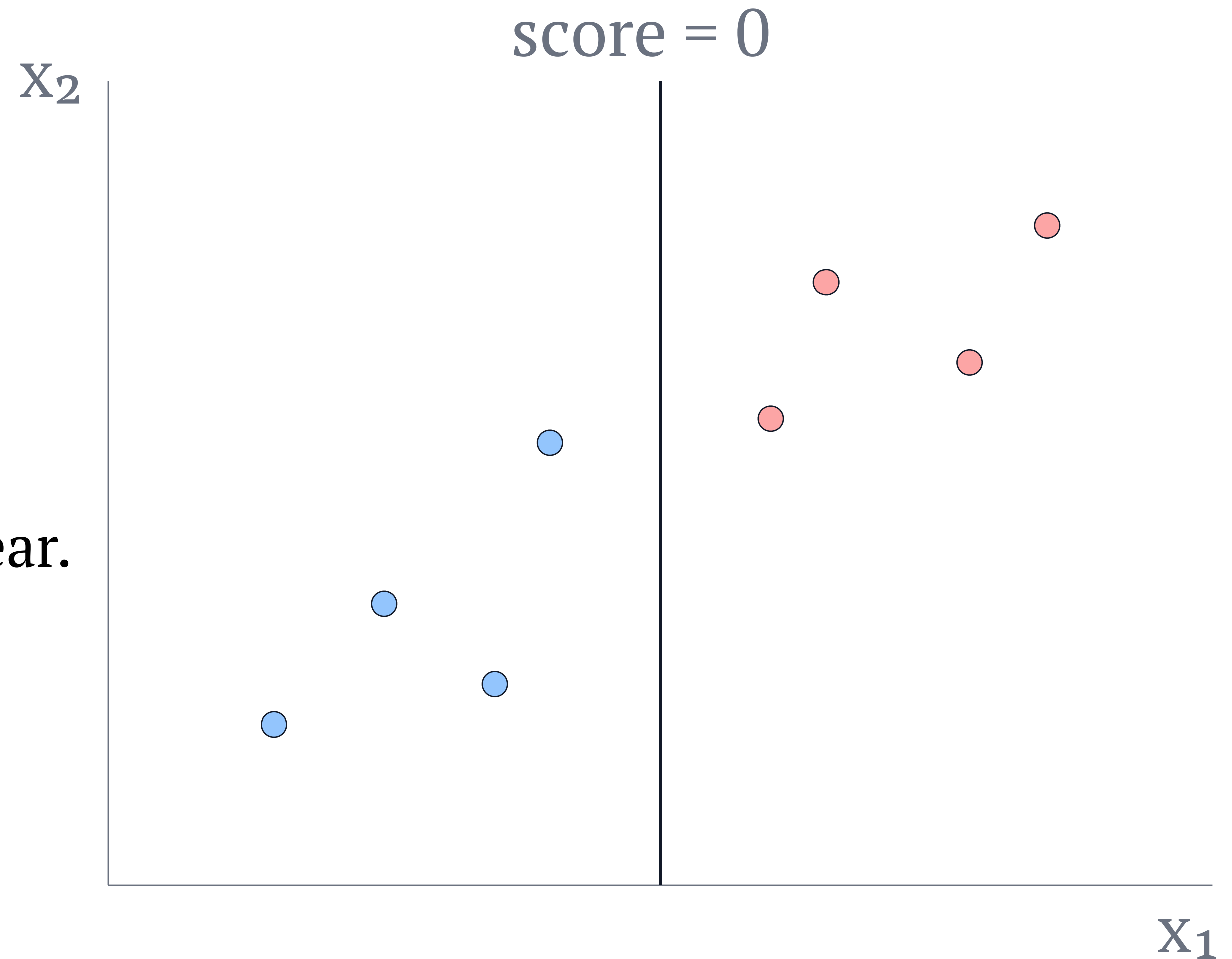
Then use gradient descent

(same optimizer story as deep
learning).

Logistic Regression

Goal: make Week 1 feel obvious before we touch MLP.

- Decision Boundary is Still Linear
 - Logistic regression adds:
 - Probabilities
 - A better loss for Classification
 - However, the boundary is still linear.



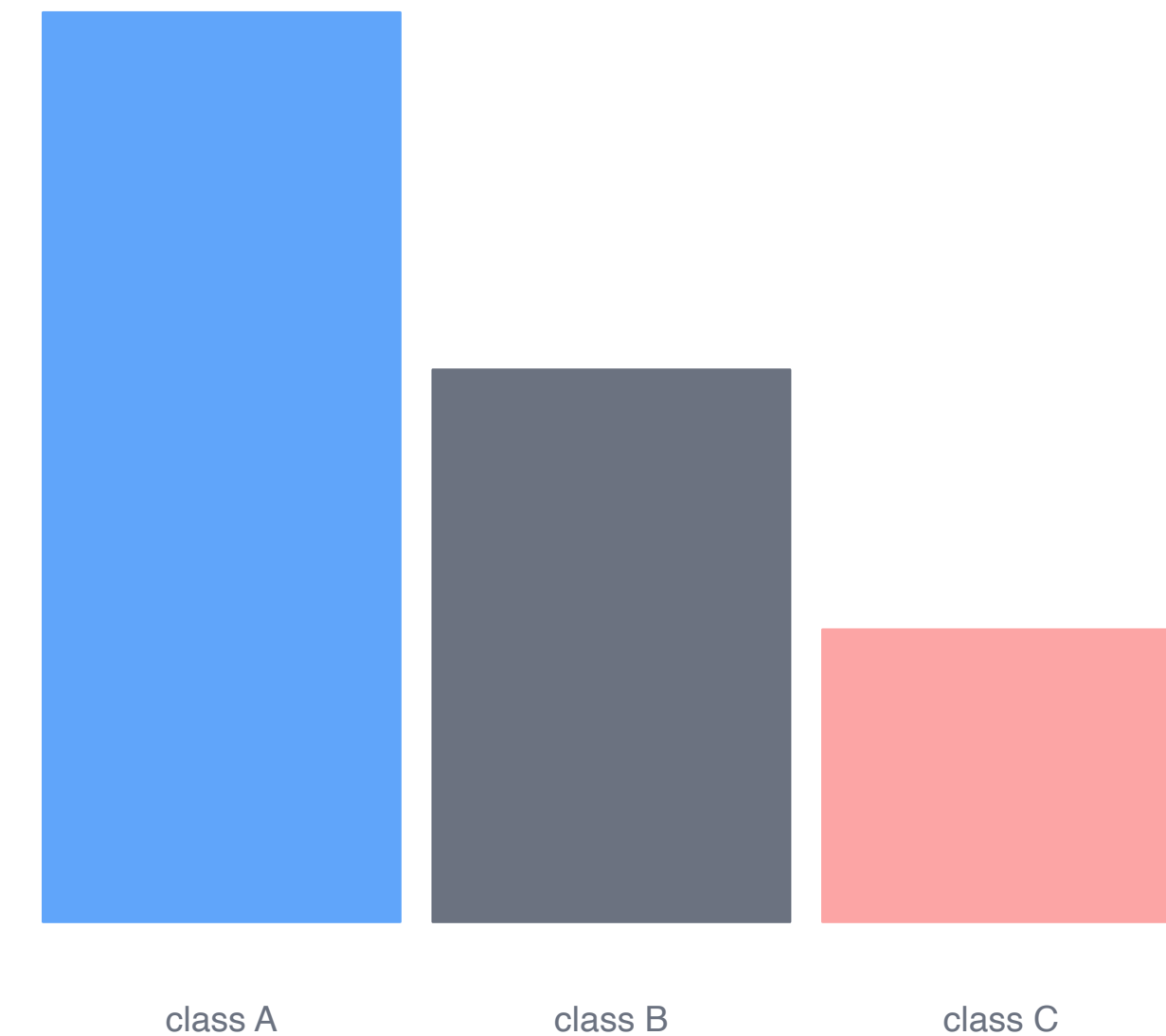
Multi Class Problems?

Goal: make Week 1 feel obvious before we touch MLP.

- Same Pattern Again:
 - linear scores $\rightarrow$ probability link (softmax) $\rightarrow$ cross-entropy

- $$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

softmax(z) $\rightarrow$ probability vector



Multi Class Problems?

Goal: make Week 1 feel obvious before we touch MLP.

- Properties for Softmax

- Normalized probabilities

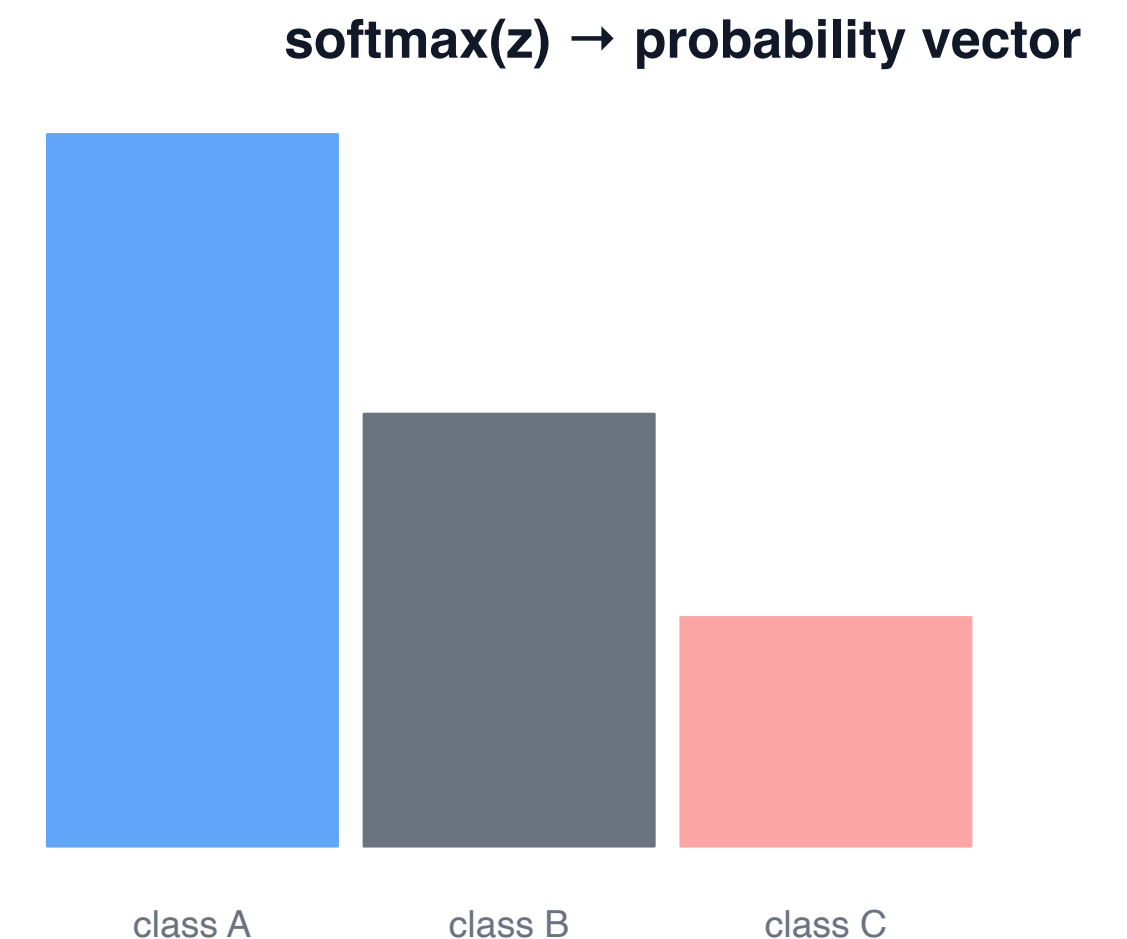
- $\sum_{i=1}^K \text{softmax}(z_i) = 1$

- Relative comparison

- $\frac{e^{z_i}}{\sum_j e^{z_j}}$, : larger score $\rightarrow$ higher probability

- Pipeline

- Linear score ($z = Wx + b$) $\rightarrow$ softmax $\rightarrow$ probability vector ($p = (p_1, p_2, \dots, p_K)$)



Multi Class Problems?

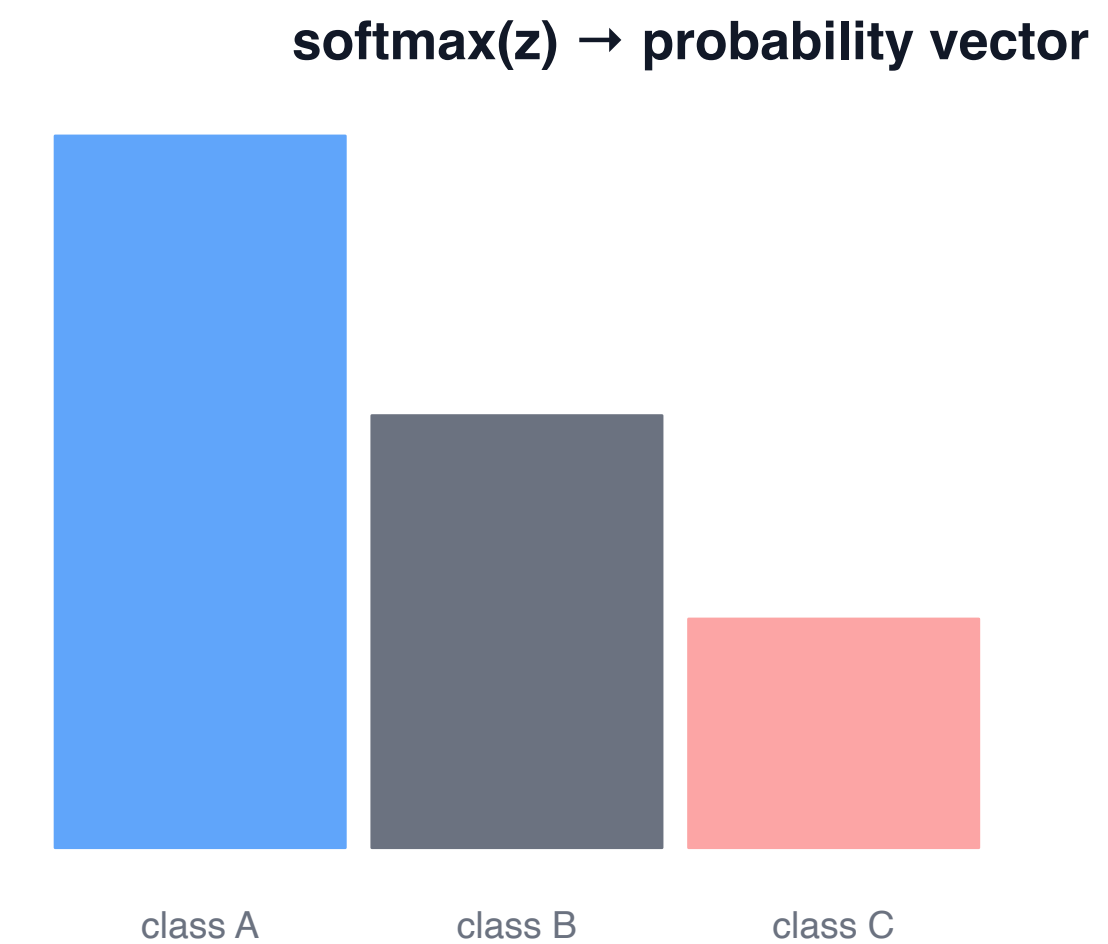
Goal: make Week 1 feel obvious before we touch MLP.

- Cross Entropy?

- $\mathcal{L} = - \sum_{i=1}^K y_i \log(\hat{y}_i)$

- For your understanding
 - Binary Cross Entropy (Logistic Regression)

- $\mathcal{L} = - \sum_{i=1}^K y_i \log(\hat{y}_i)$



Perceptron

Goal: make Week 1 feel obvious before we touch MLP.

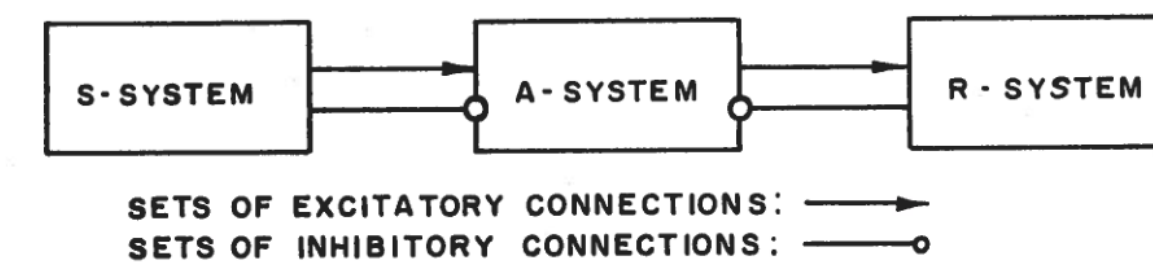
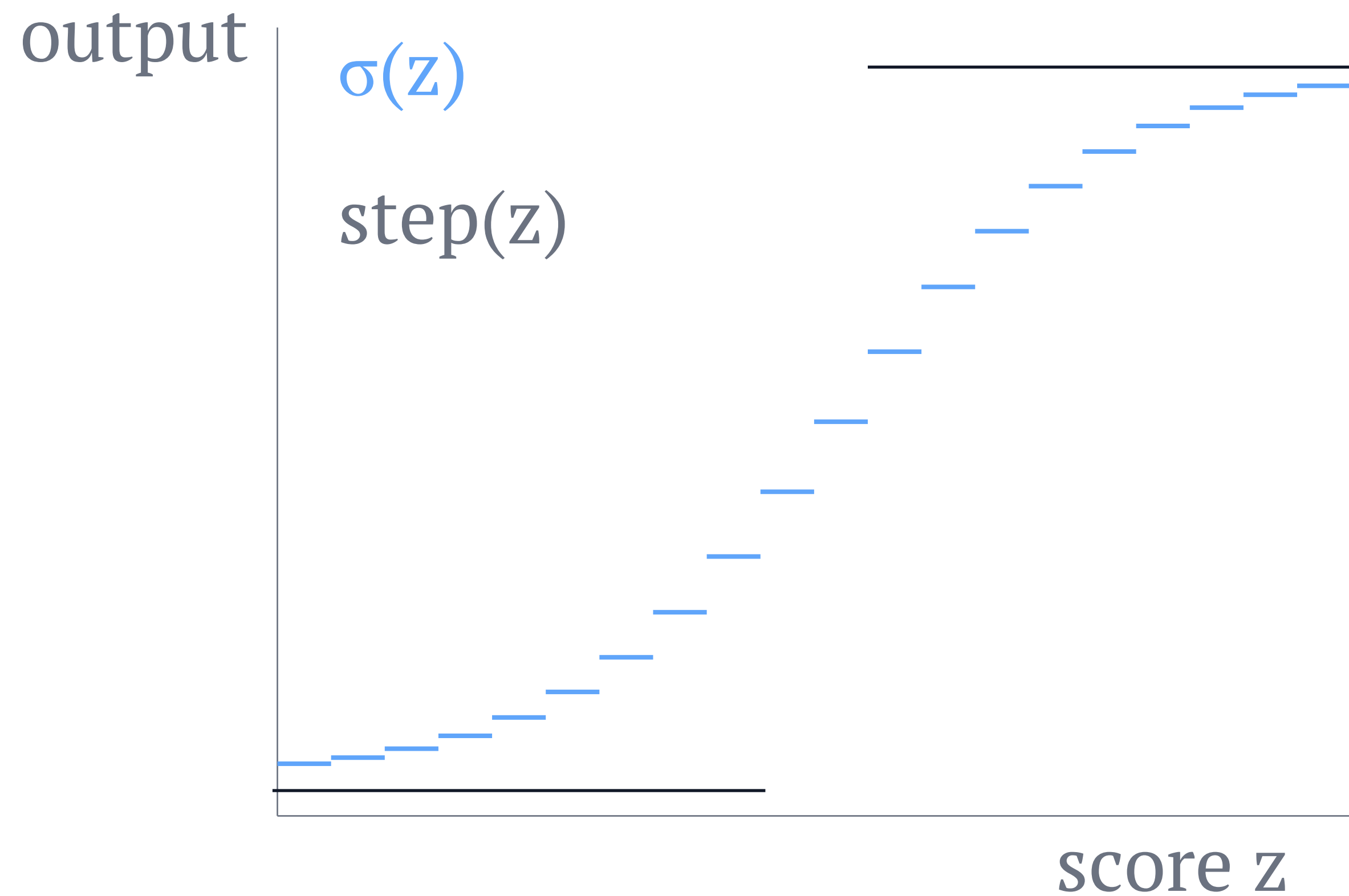


FIGURE 1
GENERAL ORGANIZATION OF THE PERCEPTRON

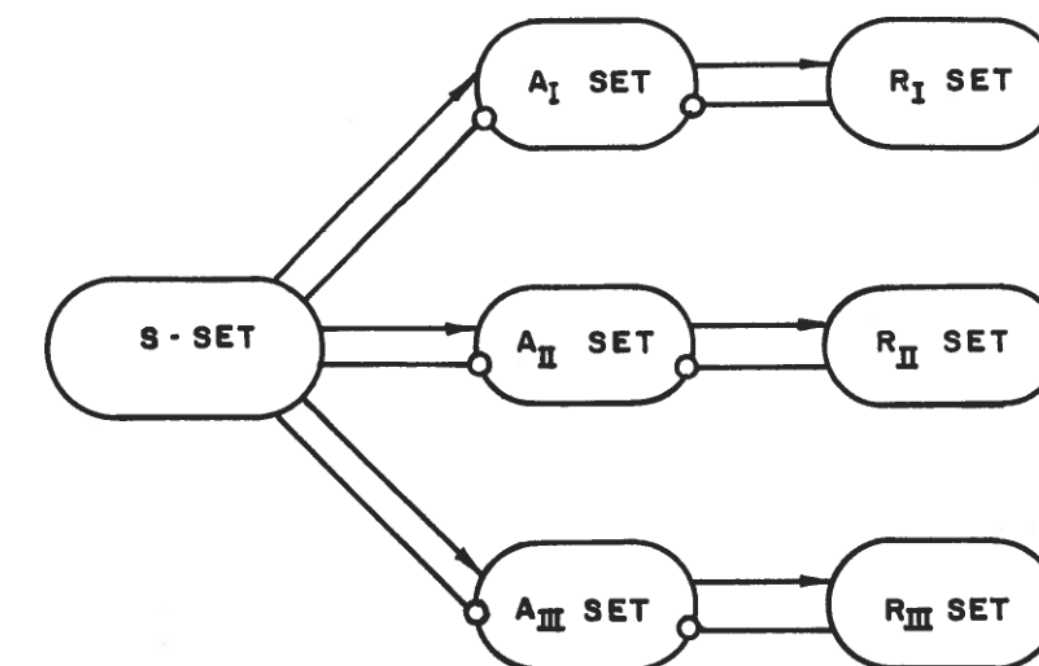


FIGURE 2
ORGANIZATION OF A PERCEPTRON WITH
THREE INDEPENDENT OUTPUT-SETS

Rosenblatt (1957): “organization

Q&A

Any question?